Telco Customer Churn

Final Analytics Report

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1. Executive Summary

This report analyses **7,043** telecom customers to understand why customers leave. Overall churn is **26.54%** (1,869 customers), putting about **$139,131** monthly revenue at risk.

Headline findings

- Contract type is the strongest driver: month-to-month churn is 42.7% vs 2.8% on two-year contracts.
- Fiber-optic customers churn at 41.9% (vs 7.4% with no internet), and electronic-check payers churn at 45.3%.
- Logistic regression captures 78.6% of churners with ROC-AUC 0.841.
- Highest risk segment churns at 48.0%.

Top recommendation: Incentivize long-term contracts and migrate high-risk customers from electronic check to automatic payment methods.

2. Problem Statement & Objectives

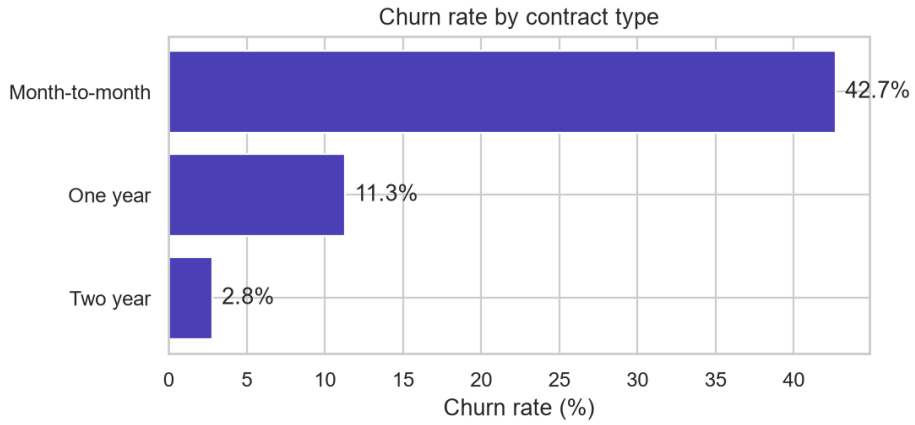
Customer churn causes revenue loss. The objective is to identify drivers, segment customers, and establish predictive models.

3. Data Source & Methodology

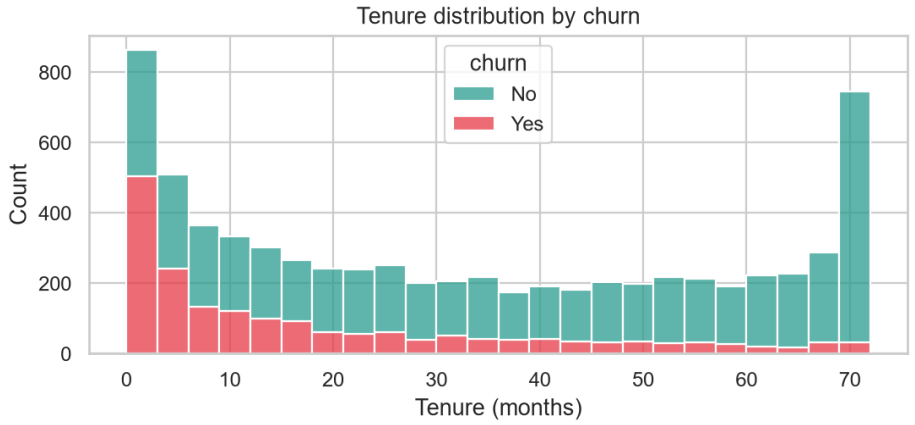
IBM Telco Customer Churn dataset with end-to-end data cleaning, SQL extraction, exploratory visualization, and machine learning models.

4. Key Findings

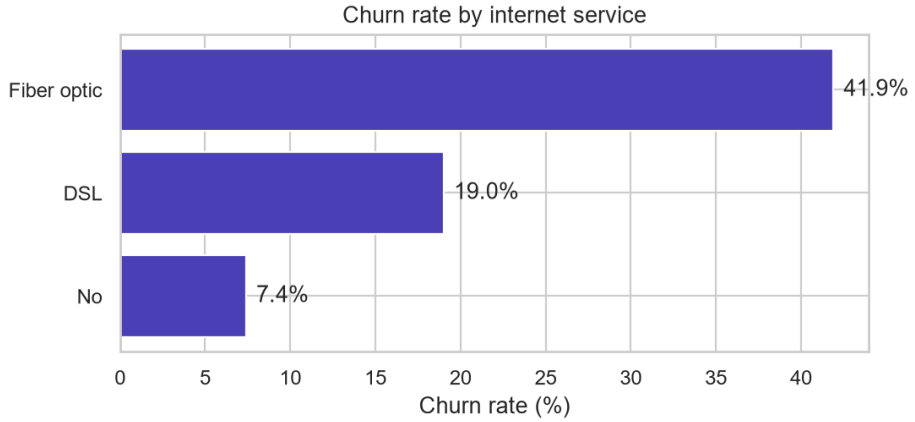
Insight 1 - Contract length:



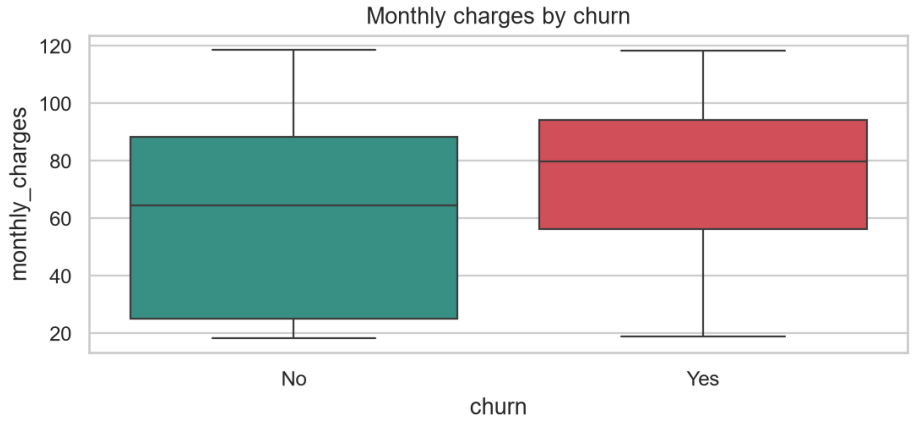
Insight 2 - Tenure:



Insight 3 - Internet Service:



Insight 4 - Monthly Charges:

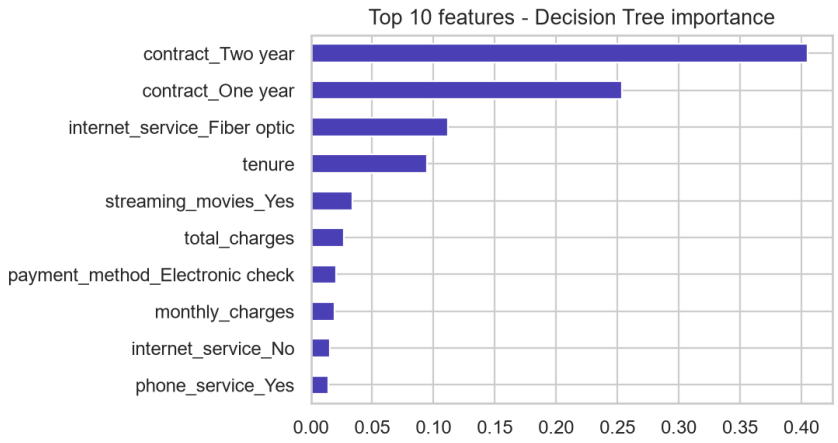


5. Statistical Analysis Results

Test	Statistic	p-value	Result
Chi-square: contract vs churn	chi2 = 1184.6 (df 2)	< 0.001	Significant
Chi-square: internet service vs churn	chi2 = 732.3 (df 2)	< 0.001	Significant
Chi-square: payment method vs churn	chi2 = 648.1 (df 3)	< 0.001	Significant
Welch t-test: monthly charges	t = 18.41	< 0.001	Significant
Welch t-test: tenure	t = -34.82	< 0.001	Significant

Predictive Models

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.74	0.507	0.786	0.616	0.841
Decision Tree	0.735	0.5	0.807	0.618	0.831



Customer Segments

Segment	Customers	Avg tenure (mo)	Avg monthly ($)	Churn rate
Segment 0	1,703	10.2	31.8	25.0%
Segment 1	1,904	59.5	93.3	15.0%
Segment 2	2,276	15.4	80.8	48.0%
Segment 3	1,160	53.6	34.9	5.0%

6. Dashboard Screenshots

[Power BI / Tableau Dashboard Screenshot Placeholder]

7. Recommendations

- Provide upgrade discounts to transition month-to-month contracts to 1-year commitments.
- Proactive outreach during the first 6 months to address customer experience bottlenecks.
- Promote paperless billing and auto-pay adoption.

8. Future Scope & Limitations

- Integrate dynamic streaming logs for real-time inference.
- Tune hyperparameter search with XGBoost for enhanced model precision.